

Week-14-Assessment-AZURE VPN

Step 1: Create Resource Group

Microsoft Azure

Search resources, services, and docs (G+I)

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lovelysh2604@gmail.com

Home > Resource Manager | Resource groups

Create a resource group

Basics

Tags

Review + create

Automation Link

Basics

Subscription: Azure for Students

Resource group name: rg-vnet-vpn-lab

Region: Central India

Tags: None

Previous

Next

Create

Microsoft Azure

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Home > Resource Manager

Resource Manager | Resource groups

Export resource groups using Bicep or Terraform

How to manage changes with deployment tools?

Summarize my costs by service

Search

+ Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

Add to service group

Group by none

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

Filter for any field...

Subscription equals all

Location equals all

Add filter

Name ↑	Subscription	Location
rg-vnet-vpn-lab	Azure for Students	Central India
VisualStudioOnline-SA69DF2076754F0C905770353AF8A5C9	Azure for Students	South India

Step 2: Create VNet-1

Microsoft Azure

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DEFAULT DIRECTORY & LOVELYSH...

Home > Network foundation | Virtual networks

Create virtual network

Basics Security Address space Tags Review + create

Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation.

[Learn more.](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure for Students

Resource group *

rg-vnet-vpn-lab

[Create new](#)

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Give feedback

Microsoft Azure

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Home

vnet1-1775881601693 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : vnet1-1775881601693

Subscription : Azure for Students

Resource group : rg-vnet-vpn-lab

Start time : 4/11/2026, 9:56:48 AM

Correlation ID : 729ceb83-89bf-4774-ab71-c2c26b16d313

> Deployment details

> Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

[Set up cost alerts >](#)

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Search

21°C

97

ENG 9:57 AM

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Home > vnet1-1775881601693 | Overview > vnet1

vnet1 | Subnets

Virtual network

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

+ Subnet

Refresh

Manage users

Delete

Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
☐	subnet1	10.0.1.0/24	-	251	-	-	-

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vnet1 | Subnets

Virtual network

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

+ Subnet

Refresh

Manage users

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name ↑	IPv4
☐	subnet1	10.0.1.0/24

Showing 1 subnet

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose

Virtual Network Gateway

Name *

GatewaySubnet

IPv4

Include an IPv4 address space

☒

IPv4 address range

10.0.0.0/16

10.0.0.0 - 10.0.255.255

Starting address *

10.0.255.0

▼ ▲

Size

/27 (32 addresses)

▼ ▲

Subnet address range

10.0.255.0 - 10.0.255.31

IPv6

Include an IPv6 address space

☐ This virtual network has no IPv6 address ranges.

Private subnet

☐

Add

Cancel

[Give feedback](#)

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Home > vnet1-1775881601693 | Overview > vnet1

vnet1 | Subnets

Virtual network

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

+ Subnet

Refresh

Manage users

Delete

Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
☐	subnet1	10.0.1.0/24	-	251	-	-	-
☐	GatewaySubnet	10.0.255.0/27	-	availability ...	-	-	-

Showing 2 subnets

[Give feedback](#)

Step 3: Create VNet-2

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information for 'lovelysh2604@gmail.com'. The main content area shows the 'vnet2-1775882095086 | Overview' page, indicating that the deployment is complete. A 'Deployment succeeded' notification is visible in the top right corner. The left sidebar lists various Azure services, with 'Subnets' selected under the 'vnet2' virtual network. The 'Add a subnet' dialog is open, showing configuration options for a new subnet named 'subnet1'. The dialog includes fields for 'Subnet purpose' (Default), 'Name' (subnet1), 'IP version' (IPv4), 'Include an IPv4 address space' (checked), 'IPv4 address range' (10.1.0.0/16), 'Starting address' (10.1.1.0), 'Size' (/24 (256 addresses)), and 'Subnet address range' (10.1.1.0 - 10.1.1.255). The 'Private subnet' section is also visible, with a checkbox for 'Include an IPv6 address space' (unchecked).

Microsoft Azure

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Home

vnet2-1775882095086 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : vnet2-1775882095086

Subscription : Azure for Students

Resource group : rg-vnet-vpn-lab

Start time : 4/11/2026, 10:05:02 AM

Correlation ID : e297308d-bbc4-4f65-bab1-e0a3388173f4

Deployment details

Next steps

Go to resource

Deployment succeeded

Deployment 'vnet2-1775882095086' to resource group 'rg-vnet-vpn-lab' was successful.

Go to resource

Go to resource group

Cost management

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Home > vnet2-1775882095086 | Overview > vnet2

vnet2 | Subnets

Virtual network

Search

+ Subnet Refresh Manage users

Create subnets to segment the virtual network address space into smaller subnets.

Search subnets

No items found

No subnets

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose

Default

Name *

subnet1

IPv4

Include an IPv4 address space

IPv4 address range

10.1.0.0/16

10.1.0.0 - 10.1.255.255

Starting address *

10.1.1.0

Size

/24 (256 addresses)

Subnet address range

10.1.1.0 - 10.1.1.255

IPv6

Include an IPv6 address space

This virtual network has no IPv6 address ranges.

Private subnet

Add Cancel

Give feedback

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Search resources, services, and docs (G+/I)

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Home > vnet2-1775882095086 | Overview > vnet2

vnet2 | Subnets

Virtual network

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

+ Subnet

Refresh

Manage users

Delete

Export to CSV

Search subnets

	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table	
☐	subnet1	10.1.1.0/24	-	251	-	-	-	
☐	GatewaySubnet	10.1.255.0/27	-	availability ...	-	-	-	

Showing 2 subnets

Give feedback

Successfully added subnet

Successfully added subnet 'GatewaySubnet' to virtual network 'vnet2'.

Step 4: Create VPN Gateway for VNet-1

Microsoft Azure

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Home

Create virtual network gateway

Basics

Tags

Review + create

Azure has provided a planning and design guide to help you configure the various VPN gateway options. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure for Students

Resource group

rg-vnet-vpn-lab (derived from virtual network's resource group)

Instance details

Name *

vnet1-gateway

Region *

Central India

[Deploy to an Azure Extended Zone](#)

Gateway Configuration

Gateway type *

☒ VPN ☐ ExpressRoute

Review + create

Previous

Next: Tags >

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Step 5: Create VPN Gateway for VNet-2

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Home

Microsoft.VirtualNetworkGateway-2026041115414 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : Microsoft.VirtualNetworkGateway-2026041111... Start time : 4/11/2026, 11:54:56 AM
Subscription : Azure for Students Correlation ID : 6fc14e24-d19f-47e2-878a-e215f53120ab
Resource group : rg-vnet-vpn-lab

Deployment details

Resource	Type	Status	Operation details
vnet2-gateway	Virtual network gateway	Created	Operation details
pip-vnet2-gw	Public IP address	OK	Operation details

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Home

CreateConnection-20260411123225 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : CreateConnection-20260411123225 Start time : 4/11/2026, 12:34:26 PM
Subscription : Azure for Students Correlation ID : d2901029-a2f3-4c9f-864a-710b77913ad2
Resource group : rg-vnet-vpn-lab

Deployment details

Resource	Type	Status	Operation details
vnet1-to-vnet2	Connection	Created	Operation details

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Step 6: Create Connection (VNet1 → VNet2)

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'Overview' page for a deployment named 'CreateConnection-20260411123732'. The deployment is in progress, indicated by a green checkmark and the text 'Deployment is in progress'. The deployment details show the deployment name, subscription (Azure for Students), resource group (rg-vnet-vpn-lab), start time (4/11/2026, 12:38:42 PM), and correlation ID (a6a00e68-1d5a-42b3-83ad-71b861f7a143). A table lists the resources created, showing a connection named 'vnet2-to-vnet1' with a status of 'Created'. On the right side, there are links for Microsoft Defender for Cloud, Free Microsoft tutorials, and Work with an expert.

Home

CreateConnection-20260411123732 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : CreateConnection-20260411123732 Start time : 4/11/2026, 12:38:42 PM

Subscription : Azure for Students Correlation ID : a6a00e68-1d5a-42b3-83ad-71b861f7a143

Resource group : rg-vnet-vpn-lab

Deployment details

Resource	Type	Status	Operation details
vnet2-to-vnet1	Connection	Created	Operation details

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STEP 7: Reverse Connection

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'Overview' page for a deployment named 'CreateVm-canonical.ubuntu-24_04-lts-server-20260411124137'. The deployment is complete, indicated by a green checkmark and the text 'Your deployment is complete'. The deployment details show the deployment name, subscription (Azure for Students), resource group (rg-vnet-vpn-lab), start time (4/11/2026, 12:44:24 PM), and correlation ID (5e3f5f59-a9ab-4c38-8ae1-0232a7294288). A section titled 'Next steps' provides recommendations for setting up auto-shutdown, monitoring VM health, and running a script inside the virtual machine. On the right side, there's a 'Deployment succeeded' notification and a 'Cost management' section.

Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260411124137 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-2... Start time : 4/11/2026, 12:44:24 PM

Subscription : Azure for Students Correlation ID : 5e3f5f59-a9ab-4c38-8ae1-0232a7294288

Resource group : rg-vnet-vpn-lab

Deployment details

Next steps

Set up auto-shutdown Recommended

Monitor VM health, performance, and network dependencies Recommended

Run a script inside the virtual machine Recommended

Go to resource Create another VM Scale out your VM

Deployment succeeded

Deployment 'CreateVm-canonical.ubuntu-24_04-lts-server-20260411124137' to resource group 'rg-vnet-vpn-lab' was successful.

Go to resource Go to resource group

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

STEP 8: Create Test VMs

VM1 in VNet1 and VM2 in VNet2

This screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'Overview' page for a deployment named 'CreateVm-canonical.ubuntu-24_04-lts-server-20260411124137'. The deployment is marked as 'Complete'. A notification banner at the top right states 'Deployment succeeded'. The left sidebar contains links for Overview, Inputs, Outputs, and Template. The main content area includes a search bar, action buttons (Delete, Cancel, Redeploy, Download, Refresh), and a summary of deployment details. Below this, there are 'Next steps' recommendations and buttons to 'Go to resource', 'Create another VM', and 'Scale out your VM'. A 'Cost management' section is visible on the right side.

Home

Microsoft Azure

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Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260411124137 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-2... Start time : 4/11/2026, 12:44:24 PM

Subscription : Azure for Students Correlation ID : 5e3f5f59-a9ab-4c38-8ae1-0232a7294288

Resource group : rg-vnet-vpn-lab

> Deployment details

✓ Next steps

Set up auto-shutdown Recommended

Monitor VM health, performance, and network dependencies Recommended

Run a script inside the virtual machine Recommended

Go to resource Create another VM Scale out your VM

Deployment succeeded

Deployment 'CreateVm-canonical.ubuntu-24_04-lts-server-20260411124137' to resource group 'rg-vnet-vpn-lab' was successful.

Go to resource Go to resource group

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Add or remove features for navigation Ctrl + Shift + F

This screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'Overview' page for a deployment named 'CreateVm-canonical.ubuntu-24_04-lts-server-20260411124652'. The deployment is marked as 'In progress'. The left sidebar contains links for Overview, Inputs, Outputs, and Template. The main content area includes a search bar, action buttons (Delete, Cancel, Redeploy, Download, Refresh), and a summary of deployment details. Below this, there are 'Deployment details' showing a table of resources and their status. A 'Next steps' section is also visible.

Home

Microsoft Azure

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Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260411124652 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-2... Start time : 4/11/2026, 12:49:09 PM

Subscription : Azure for Students Correlation ID : 2a1f8d4b-df66-4ef6-81ca-981267a41aea

Resource group : rg-vnet-vpn-lab

> Deployment details

Resource	Type	Status	Operation details
vm2594_x2	Microsoft.Network/networkInter	Created	Operation details
vm2-nsg	Network security group	OK	Operation details

> Next steps

Microsoft Azure

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Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260411124652 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-2... Start time : 4/11/2026, 12:49:10 PM
Subscription : Azure for Students Correlation ID : 2a1f8d4b-df66-4ef6-81ca-981267a41aea
Resource group : rg-vnet-vpn-lab

Deployment details

Next steps

Set up auto-shutdown Recommended
Monitor VM health, performance, and network dependencies Recommended
Run a script inside the virtual machine Recommended

Go to resource Create another VM Scale out your VM

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

STEP 9: Allow Ping

Home > vm1 | Network settings > vm1860_z2

vm1860_z2 | IP configurations

Network interface

Virtual network: vnet1

Subnet *: subnet1 (10.0.1.0/24) 250

Private and public IP addresses can be assigned to a virtual machine's private and public IPv4 addresses as necessary to a network interface. IP address prefix types can only be created as a secondary IP configuration.

+ Add Make primary Delete

Name	IP Version	Type
☐ ipconfig1	IPv4	Primary

Apply Discard changes

Edit IP configuration

vm1860_z2

Private IP address settings

Add a public IP address

Name * pip-vm1

SKU Standard

Static IPs are assigned at the time the resource is created and released when the resource is deleted.

Assignment Static

OK Cancel

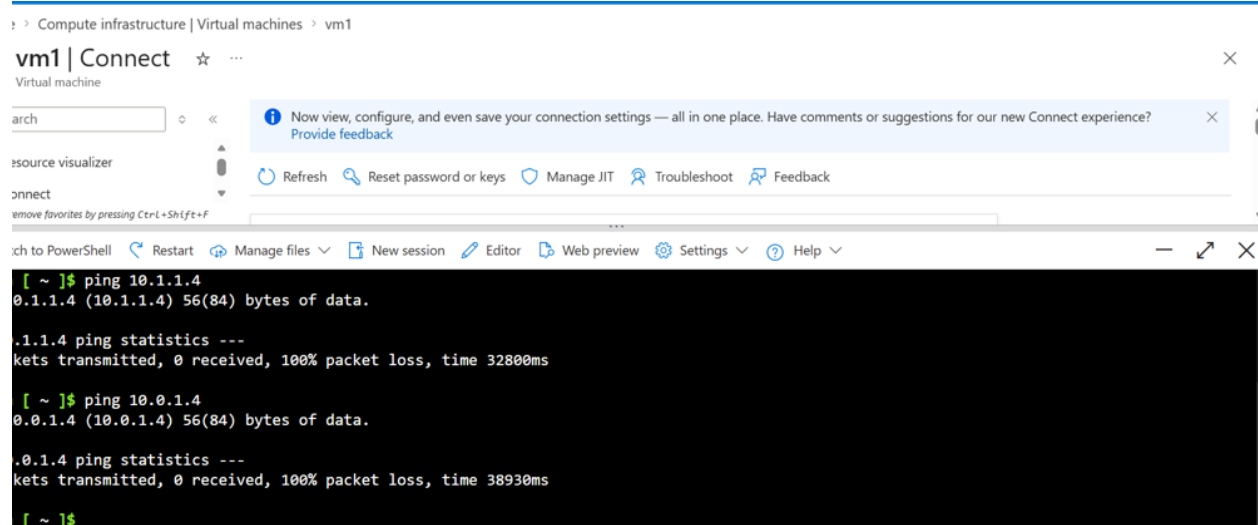
Add a public IP address

Create a public IP address

Save Cancel

Give feedback

Step 10: Test Connectivity



The screenshot shows the 'vm1 | Connect' page in the Azure portal. The page title is 'vm1 | Connect' with a star icon and a close button. Below the title, it says 'Virtual machine'. There is a search bar with 'arch' entered. A notification banner at the top right says: 'Now view, configure, and even save your connection settings — all in one place. Have comments or suggestions for our new Connect experience? Provide feedback'. Below the notification, there are links: 'Refresh', 'Reset password or keys', 'Manage JIT', 'Troubleshoot', and 'Feedback'. The main content area is a terminal window titled 'Connect to PowerShell'. The terminal shows the following commands and output:

```
[ ~ ]$ ping 10.1.1.4
0.1.1.4 (10.1.1.4) 56(84) bytes of data.

.1.1.4 ping statistics ---
0 packets transmitted, 0 received, 100% packet loss, time 32800ms

[ ~ ]$ ping 10.0.1.4
0.0.1.4 (10.0.1.4) 56(84) bytes of data.

.0.1.4 ping statistics ---
0 packets transmitted, 0 received, 100% packet loss, time 38930ms

[ ~ ]$
```